ARPIT PATEL

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Portfolio: https://arpit-patel20.github.io/my_portfolio | ArcGIS StoryMaps: https://arcg.is/1buzDn

PROFESSIONAL SUMMARY

GIS Analyst with a strong foundation in geospatial data management, spatial analysis, and cartographic production using ArcGIS Pro and ArcGIS Online. Experienced in working with multidisciplinary datasets, performing QA/QC, and developing GIS solutions to support environmental and infrastructure-related projects. Skilled in data conversion, geodatabase management, and creating high-quality maps, figures, and web GIS applications. Demonstrates strong analytical thinking, attention to detail, and the ability to communicate complex spatial information clearly within collaborative project environments.

TECHNICAL SKILLS

GIS Platforms: ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise (concepts), Experience Builder, Dashboards, StoryMaps

Data & Analysis: Geodatabases, spatial data management, QA/QC, data validation, coordinate systems, geoprocessing, raster analysis (NDVI, Tasseled Cap), overlay analysis

Data Conversion & Integration: CAD (DWG) data handling, tabular (CSV/Excel) to spatial conversion, raster/vector integration, schema alignment

Automation & Tools: Python (ArcPy), ModelBuilder, custom tool development, workflow automation

Cartography & Output: Map layouts, figures, technical maps, report graphics, visualization design

GIS PROJECT EXPERIENCE

Wildfire Risk Assessment & GIS Analysis Project

Client: Centre for Innovation and Research in Unmanned Systems (CIRUS Lab), SAIT

- Contributed to a multidisciplinary environmental project, integrating GIS, remote sensing, and terrain analysis to support wildfire risk assessment
- Collected, organized, and converted datasets from multiple sources including satellite imagery, DEM, and vector datasets into structured GIS formats
- Built and maintained datasets within a geodatabase, ensuring consistency, usability, and proper schema design
- Performed QA/QC and data validation, including attribute verification, geometry correction, and coordinate system standardization
- Conducted advanced raster analysis using NDVI and Tasseled Cap transformations to assess vegetation condition and land cover change
- Applied supervised classification techniques and performed accuracy assessment using confusion matrices, ensuring reliable and high-quality classification results
- Performed raster modelling (reclassification and weighted overlay) to support environmental risk assessment and decision-making
- Created high-quality cartographic outputs (maps, figures, and visualizations) for technical reporting and presentations
- Developed web GIS applications using ArcGIS Online and Experience Builder, enabling stakeholders to explore spatial data interactively
- Designed and configured a mobile mapping application using ArcGIS Field Maps to support first responders, enabling field data access, real-time updates, and improved situational awareness
- Developed a custom ArcGIS tool using ArcPy and ModelBuilder to automate the classification and analysis workflow, improving efficiency, consistency, and repeatability of results
- Used ArcPy scripting to streamline geoprocessing tasks, reducing manual effort and enhancing workflow performance
- Documented workflows, datasets, and processes to ensure reproducibility and alignment with professional GIS standards

Chicago Arson Spatial Analysis, GIS Data Management & Network Analysis

- Designed and implemented a relational geodatabase to integrate arson incident data with community boundary datasets, ensuring structured data organization and efficient spatial analysis
- Collected and converted raw datasets (CSV and tabular formats) into spatial feature classes, ensuring accurate geocoding, coordinate system alignment, and spatial referencing
- Performed comprehensive data cleaning, QA/QC, and validation, including removal of null values, geometry repair, attribute consistency checks, and projection standardization
- Conducted spatial aggregation using Spatial Join to summarize incident data at the community area level for analysis
- Applied spatial statistical analysis including Hotspot Analysis (Getis-Ord Gi*), Cluster and Outlier Analysis (Anselin Local Moran's I), and Global Moran's I to identify spatial clustering patterns and assess statistical significance
- Performed network analysis using ArcGIS Network Analyst, generating service area models to evaluate emergency response coverage and accessibility of fire stations relative to identified hotspots
- Developed and automated GIS workflows using ModelBuilder, improving efficiency and repeatability of geoprocessing tasks

- Created high-quality cartographic outputs, including thematic maps, analytical figures, and layout designs adhering to cartographic standards
- Published results through an ArcGIS StoryMap, integrating maps, charts, and narrative text to effectively communicate findings to both technical and non-technical audiences
- Organized datasets, outputs, and workflows to support scalability, reproducibility, and professional GIS project standards

PROFESSIONAL EXPERIENCE

Lupin Ltd. **India**
Trainee Officer – Environmental *Nov 2024 – Aug 2025*

- Collected and validated environmental datasets ensuring accuracy, consistency, and completeness
- Performed data QA/QC and supported reporting workflows
- Prepared technical reports and maintained structured documentation
- Collaborated with team members to support operational and compliance activities

LEADERSHIP EXPERIENCE

Vice President — BGIS Student Club **Calgary, Alberta**
Southern Alberta Institute of Technology *Sept 2025 – Present*

- Organized geospatial networking events connecting students with GIS professionals and industry partners including Esri Canada, AltaLIS, and Stantec
- Coordinated the BGIS Geospatial Industry Night event bringing together students and industry leaders
- Collaborated with team members to manage event logistics, communication, and stakeholder coordination in a professional environment.

EDUCATION

Southern Alberta Institute of Technology (SAIT)
Bachelor of Applied Technology – Geographic Information Systems (GIS)
Calgary, Alberta | Currently enrolled

Relevant Studies:

- Spatial analysis and geoprocessing using ArcGIS Pro
- Geodatabase design and spatial data management
- Cartographic design and professional map production
- Python scripting and ModelBuilder automation
- Web GIS development using ArcGIS Online and Experience Builder

CVM University
Master of Environmental Science and Technology
India | 2023

Focus Areas:

- Environmental impact assessment
- Environmental monitoring and regulatory compliance
- Technical report writing and scientific documentation